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Mesoporous calcium silicate glasses. I. Synthesis

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Abstract

The conventional method of preparing glasses is quenching a melt. However, as with many multicomponent glasses, only a certain range of CaO–SiO₂ compositions can be melt derived. Sol–gel processing allows one to produce gel-glasses with compositions lying within the liquid–liquid immiscibility dome of the CaO–SiO₂ system. Crack-free silica–calcium xerogel monoliths of various shapes and sizes of compositions expressed by the following formula $x\text{CaO} \cdot (1-x)\text{SiO}_2$, where $0 \leq x \leq 0.5$ mole fraction, were prepared via the sol–gel technique, starting from tetraethyl orthosilicate and calcium nitrate. In this paper the processing technique is described. The materials produced were characterised using X-ray diffraction, infrared (FTIR) spectroscopy, energy dispersive scanning electron microscopy (SEM-EDS) as well as differential thermal analysis (DTA). The gel-glasses produced are confirmed to be amorphous even after stabilisation at 600 °C, with crystallisation temperatures above 850 °C. When sintered at 1000 °C they form crystalline phases in accordance to the CaO–SiO₂ phase diagram. Acid digestion analysis confirms that the compositions of the gel-glasses are similar to that of the computed values. The gel-glasses are homogeneous throughout the monolith.

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1. Introduction

The sol–gel process is a chemical synthesis technique for the preparation of glasses and ceramics. The chemistry involved in the process is based on inorganic polymerisation reactions of metal alkoxide precursors $\text{M}(\text{OR})_n$, where M represents the network forming element such as Si and R is an alkyl group $\text{C}_x\text{H}_{2x+1}$ [1]. These precursors undergo hydrolysis and condensation re-

actions, usually when dissolved in a solvent, to form soluble metal hydroxides, which ultimately leads to the formation of continuous metal–oxygen–metal link as in an inorganic network that spans throughout the solvent medium.

The main features of the multi-component sol–gel process are that a homogeneous solution is formed before polymerisation occurs and the processing takes place at lower temperatures than in conventional methods [2]. Thus, not only can improved chemical homogeneity be achieved with good control over the final composition and tailoring of the surface characteristics of the product but also phase separation, crystallisation and chemical decomposition can be avoided.

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The phase diagram of the CaO–SiO₂ system showing the glass compositions investigated is illustrated in Fig. 1 [3]. As can be seen from the phase diagram, at the lower end of CaO (0.98–0.72 mole fraction silica) a metastable liquid–liquid immiscibility region exists. This miscibility gap is asymmetric with a consolute temperature of 1871 °C and a solvus temperature of 1705 ± 10 °C [4]. It has been reported [4] that glasses quenched from above 1700 °C consist of a clear siliceous phase and a white opaque Ca-rich phase, which makes it impossible to prepare glasses with compositions in this region via the traditional melt quench method. On the other hand, to prepare glasses with intermediate Ca content (up to 0.5 mole fraction calcium), the melting temperatures needed to achieve a solution of starting materials is above 1600 °C [3].

The sol–gel method is particularly suitable for the preparation of glasses with compositions discussed above. The preparation of monolithic glasses, glass fibres, pure fine powders and other ceramic materials from metal alkoxides or metal

salts by the gel-method has been reported since late 1970s by several researchers [5–9]. Roy [10] suggested that because of the excess free energy of a gel compared to a glass of the same composition, a gel might be converted to glass at temperatures substantially below the liquidus temperature of that composition.

However, only a few such glasses with compositions in the calcium silicate binary system have been prepared by the sol–gel procedure. The earliest attempt by Hayashi and Saito [11] in 1980 reported an inability to produce homogeneous monoliths when a salt was used as the Ca source. They also faced difficulties with hydrolysis of alkoxides, which resulted in precipitates of the hydrolysed product. In 1992 Bansal [3] described the formation of crack-free monoliths containing up to 20-mol% CaO.

The complexity arises because, unlike the silicon alkoxides, alkoxides of Group I and II metals are usually solid, non-volatile and have very low solubility [1]. Metal salts provide a viable alter-

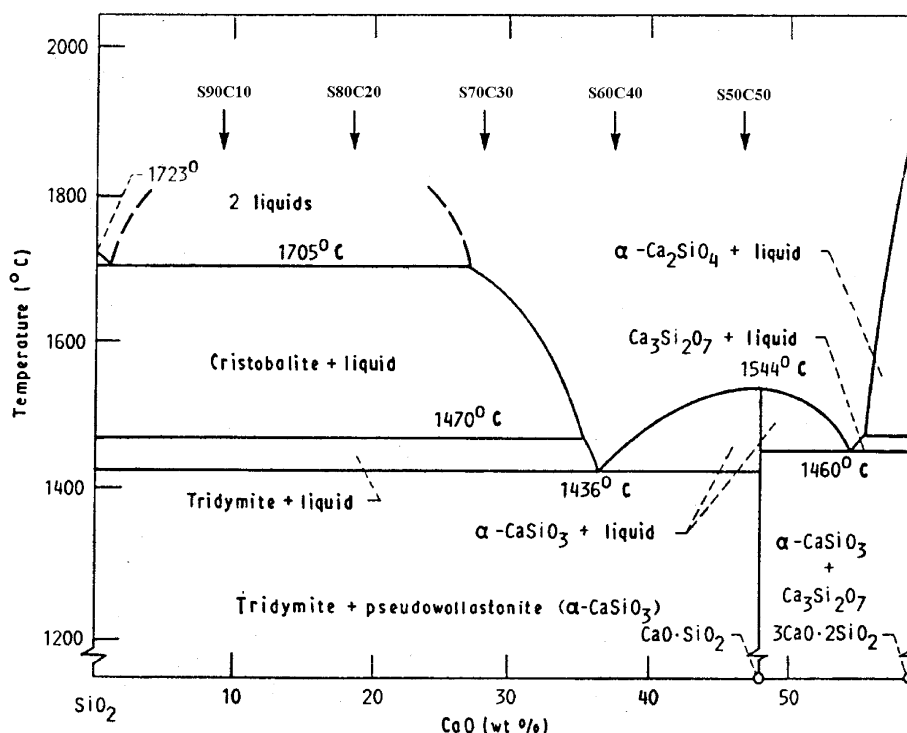


Fig. 1. Partial phase diagram of SiO₂–CaO system [3].

native and are usually in the form of nitrates as they are less thermally stable compared to chlorides or sulphates and the anion is relatively easier to remove by decomposition. Although acetate salts can also be used, they do not thermally degrade cleanly and thus leave carbonaceous residues [1]. Also acetate solutions are basic (high pH) and therefore lead to rapid gelation in silicate systems.

Hayashi and Saito [11], who first prepared gels in the binary CaO–SiO₂ system, used calcium ethylate (Ca(OEt)₄) by reacting high purity Ca metal with ethanol. They mixed both Si(OEt)₄ and Ca(OEt)₄ in dried ethanol under Ar gas. Gelation was achieved by either adding water or exposing the solution to moisture. In the former method, a translucent or opaque gel was formed in several hours whereas with the latter method a transparent or translucent gel was formed in a few days. They also observed that with the addition of water, the hydrolysis products were more likely to precipitate.

Hayashi and Saito also reported that when calcium was introduced in the form of calcium nitrate they obtained a translucent or opaque gel. At higher temperatures this process resulted in fluffy white powders containing crystalline Ca(NO₃)₂ · nH₂O [11].

Bansal [3], however, was able to successfully use calcium nitrate tetrahydrate [Ca(NO₃)₂ · 4H₂O] as the calcium source where the alcoholic solution of calcium nitrate was mixed with tetraethyl orthosilicate (TEOS), which had also been diluted with ethanol. Water was introduced drop-wise and a clear transparent gel was obtained in less than an hour to several days depending on the reaction temperature, water concentration and calcium content. Catauro and Laudisio produced similar results [12] in 1997 for glasses of compositions 2CaO · 3SiO₂ using tetramethyl orthosilicate (TMOS) and calcium nitrate tetrahydrate as starting materials.

Catauro and Laudisio [13] also used calcium acetate as the cation precursor. They used a two-step process where during the first step TMOS (diluted in dimethylsulphoxide) was partially hydrolysed using half the theoretical amount of water and acetic acid as a catalyst. In step two, the

Table 1
Nominal compositions of gel-glasses produced

Acronym	SiO ₂ (mol%)	CaO (mol%)	SiO ₂ (wt%)	CaO (wt%)
S90C10	90	10	90.62	9.38
S80C20	80	20	81.08	18.91
S70C30	70	30	71.43	28.57
S60C40	60	40	61.64	38.36
S50C50	50	50	51.72	48.28

required amount of calcium acetate was dissolved in excess water and dimethylsulphoxide. Both solutions were then mixed together slowly and gelation was achieved within 24 h.

Vallet-Regí and co-workers have recently published a series of papers on the bioactivity of binary CaO–SiO₂ sol–gel glasses [14,15]. Although TEOS and calcium nitrate tetrahydrate were used as starting materials these publications do not describe the processing of monolithic gel-glasses. After gelation and aging, the gels are dried at 150 °C for 2 days. These were then ground and the powder was compacted (at 50 MPa uniaxial and 150 MPa isostatic pressures) into discs before further sintering at 700 °C.

The aim of this work was to produce homogeneous, crack-free, monolith calcium silicate gel-glasses within the liquid–liquid immiscibility region using a simple technique, which could be easily scaled up. Gel-glasses with compositions presented in Table 1 were prepared. The compositions can be expressed by the following formula: $x\text{CaO} \cdot (1-x)\text{SiO}_2$ where $0 \leq x \leq 0.5$ mole fraction. The processing of these binary gels is also briefly mentioned in another publication in which the bioactivity of the gel-glasses in the said compositional range is discussed in detail [16].

2. Experimental

2.1. Materials

The following compounds were used for the processing of calcia–silicate gel-glasses: TEOS, Si(OC₂H₅)₄, 98% and calcium nitrate tetrahydrate, Ca(NO₃)₂ · 4H₂O, ACS reagent. Deionised (DI)

water was obtained from an instant purifier with pH 5.5 and nitric acid was used as the catalyst.

The rate of hydrolysis of TEOS is slow in neutral solutions [2] and requires an acid or base catalyst. 2N nitric acid (HNO_3) is used to catalyse the TEOS hydrolysis during the sol preparation along with molar (R) ratios $\text{H}_2\text{O}/\text{TEOS}$ of 12:1 (i.e. excess water) and volume (v) ratios $\text{H}_2\text{O}/\text{HNO}_3$ of 6:1 (i.e. molar ratio of acid/hydroxide – 0.048), as used by several investigators [2,9,17]).

2.2. Mixing order

2N HNO_3 was added to DI water and gently stirred for 5 min. TEOS was then added in small amounts over a 30-min period. This mixture is maintained for a half hour to ensure complete hydrolysis and the progression of condensation. The calcium nitrate was then added to this mixture and allowed to dissolve. Pouring and casting was achieved an hour later.

2.3. Gel synthesis

Fig. 2 schematically illustrates the process followed to obtain cylindrical glass monoliths. The sol was prepared at room temperature in Pyrex glass beakers using the mixing order described above. They were then cast into polymethyl pentene moulds of different sizes (diameter range: 10–25 mm and thickness up to 30 mm) for gelation.

Both aging and drying of wet gels were conducted in a programmable oven. Aging of the gels took place at 60 °C for 72 h. The moulds were transferred into the oven after the gelation period and the oven was programmed to heat up to 60 °C at a heating rate of 5 °C/min. The drying of the gels

Table 2
Drying schedule

Stage	Temperature (°C)	Duration (h)	Gradient (°C/min)
1	60	20	0.1
2	90	24	0.1
3	130	40	0.1

Table 3
Lindberg furnace schedule for stabilisation of gels

Stage	Ramp (°C/min)	Length (°C)	Dwell (h)
1	1.0	100	0
2	0.5	300	2
3	1.0	600	5
4	5	Rtp	0

was carried out in the same jar by loosening the screw lids to allow gas evaporation and heating the gels with a three-stage schedule listed in Table 2.

Stabilisation of the dry gels was carried out in a programmable furnace. The dried gels were allowed to cool to room temperature and then transferred to alumina plates. The stabilisation schedule used a slow ramping rate (see Table 3) to obtain crack free monoliths.

2.4. Gel characterisation

DTA curves were recorded under dynamic air-flow at a heating rate of 10 °C/min on crushed bulk specimens from room temperature to 1300 °C. An ASI thermo-analyser model was used with Al_2O_3 as the reference material.

The surfaces of the gel-glasses made were analysed using a powder diffractometer. XRD measurements were performed by the step scanning method with CuK_α radiation. The instrument was

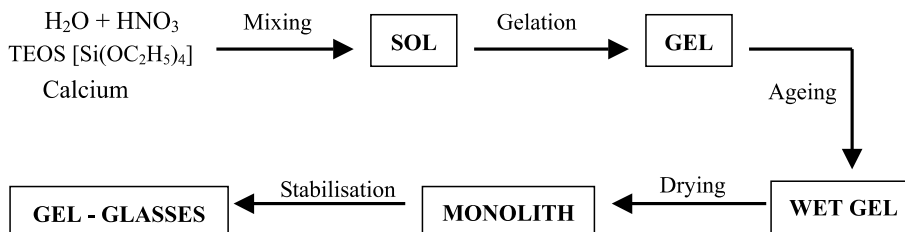


Fig. 2. Process flow chart.

operated at 40 kV and 40 mA with a 0.040° 2θ step and a count rate of 30 s/step, the 2θ values ranging from 10° to 50° . The surfaces were also monitored using a FTIR spectrometer. The spectra were recorded between 400 and 4000 cm^{-1} , measuring the diffuse radiation reflected by the surface of the bulk sample. The SEM-EDS study was performed in a 25 keV microscope. For the SEM study the specimens mounted using quick set resin, polished and was coated with a thin film of carbon.

An acid digestion technique, was performed on the gels, to analyse the composition of the final gel-glasses since the final oxide content often differs slightly from those calculated. This is due to the different processing conditions used (i.e. R ratio) for hydrolysis and condensation of the alkoxide (in the case of SiO_2 content). It is also possible for cations to leach out during the aging and drying of the gels, especially when it is carried out in the presence of the pore liquor (as described in this paper) where a pH gradient exists, resulting in a lower CaO content.

For the analysis 0.25 g of ground sample is mixed with 0.75 g of lithium metaborate and lithium tetraborate flux. This mix was then poured carefully into a platinum crucible and placed into a furnace. The ensemble is heated to 1000°C for 20 min followed by cooling to room temperature. The contents of the cooled platinum crucible are then dissolved in a solution containing 25 ml of 2 N nitric acid and 150 ml DI water. This mix is then transferred into a 250 ml volumetric flask and made up to that volume with DI water. The final diluted solution is stored in a polythene bottle until further analysis by inductive coupled plasma-atomic emission spectrometer. The elemental concentrations are then computed into wt% of the oxide with the aid of a software programme.

3. Results

3.1. Gelation

The gel point (t_{gel}) varied between 24 h for high Ca gels (0.5–0.3 mole fraction Ca) and 36 h for low Ca gels (S80C20 and S90C10 samples). The reduction in gelling time with increasing Ca^{2+} content

is thought to be due to the lowering of ionic charge on sol particles by the salt. Clear and transparent gels were obtained for all compositions.

3.2. aging and drying

At the end of the aging period shrinkage of the gels was on average found to be 10–20% of the original size, depending on size and composition of the original gel. Smaller gels with higher Ca content had higher % shrinkage. No change in the visual colouration of the gels was observed. The wet-gels after aging were free of cracks and the yield at this stage of the process was 100%.

During the drying process the discs transformed from transparent to translucent/opaque. Lower Ca content gels were white in colour (S90C10 samples) whereas the other compositions ranged from light yellow (S80C20 and S70C30 samples) to yellow (S60C40 and S50C50). This was due to the increasing nitrate content in the gels. When the wet-gels were exposed to the atmosphere the discoloration was stronger.

The yield at this stage of the process was on average 90%. The larger the gel size and higher the Ca content the lower the yield. The yield is the ratio of the number of crack free monoliths to the number of monoliths cast. Usually, a batch consisted of between 70 and 100 monoliths being cast. This number is limited only due to the size of the oven and furnace in use as well as the ease of handling by one person during the processing.

3.3. Stabilisation

With the slower program for stabilisation, more than 99% of the gels survived this final stage of the process. For the stabilised binary gels the transparency reduced with the composition and the size of the gels. For smaller specimen size, low Ca gels were clear and transparent whereas the high Ca gels were transparent to translucent. The larger gels with high Ca content were opaque.

3.4. Pyrolysis of wet binary gel

Fig. 3 is a DTA/TGA curve for a wet S50C50 gel sample. The sample was retrieved after gelation

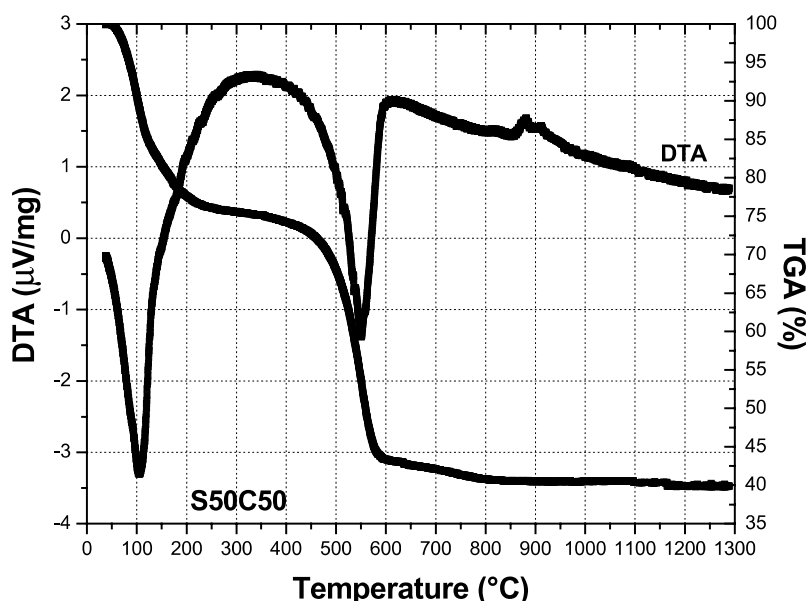


Fig. 3. DTA and TGA curves for wet S50C50 gel glass composition.

at room temperature and allowed to age at that temperature for a week before the analysis. Approximately 19 mg of the sample was placed in a platinum crucible and calcined to 1300 °C.

There is an initial endothermic process around 100 °C which can be attributed to loss of residual water and ethanol (i.e. pore liquor) corresponding to approximately 25% reduction in weight. This weight loss is more gradual and occurs over a range of temperatures up to 450 °C. This is attributed to the loss of organics (i.e. alkoxy group) after the loss of pore liquor. A second endothermic process occurs at around 550 °C with a weight loss of ~35% attributed to the loss of nitrates used in the sol preparation. No weight loss was observed after this temperature. The exothermic peak around 900 °C is attributed to the crystallisation of

CaSiO₃ (β-wollastonite) and cristobalite, as confirmed by XRD analysis (see Fig. 7).

Similar observations were made for all the other binary compositions prepared. The temperature of crystallisation increased with lower Ca content. The crystallisation temperatures are listed in Table 4.

3.5. Comparison of DTA at different stages in the process

Figs. 4 and 5 compare the DTA and TGA curves of S70C30 samples during different stages in the process. For the initial endothermic process, the heat released decreases with the progress of the processing. The aged gel, termed WET, contains the most amount of water while the STA (stabilised) samples contain hardly any water. The absence of the endothermic peak at approximately 500 °C for the stabilised sample confirms the total decomposition of nitrates and any other undesired components of the starting material. The TGA curves corresponding to the above DTA traces are seen in Fig. 5. The slight weight loss observed in the stabilised sample is attributed to structural densification.

Table 4
Crystallisation temperatures of binary gel-glasses

Sample	~Crystallisation temperature (°C)
S50C50	880 ± 12
S60C40	930 ± 8
S70C30	960 ± 5
S80C20	970 ± 5
S90C10	970 ± 8

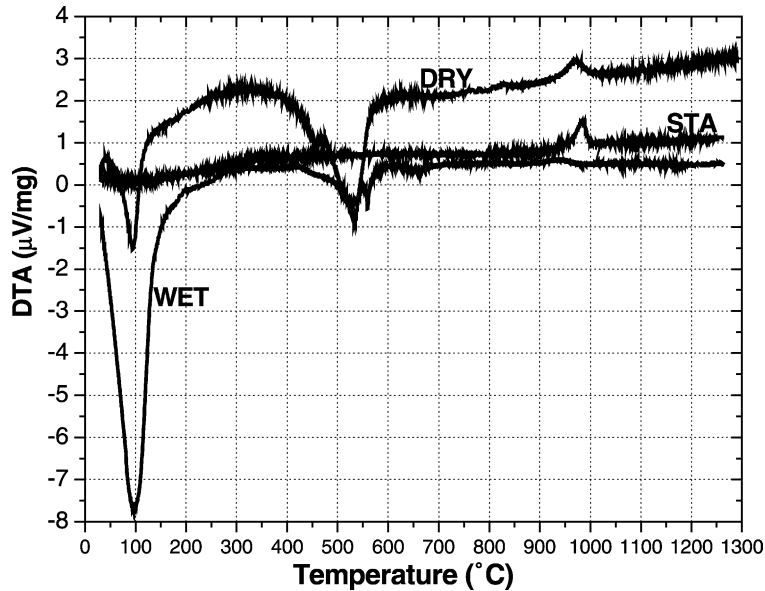


Fig. 4. DTA curves for wet (WET), dry (DRY) and stabilised (STA) S70C30 gel composition.

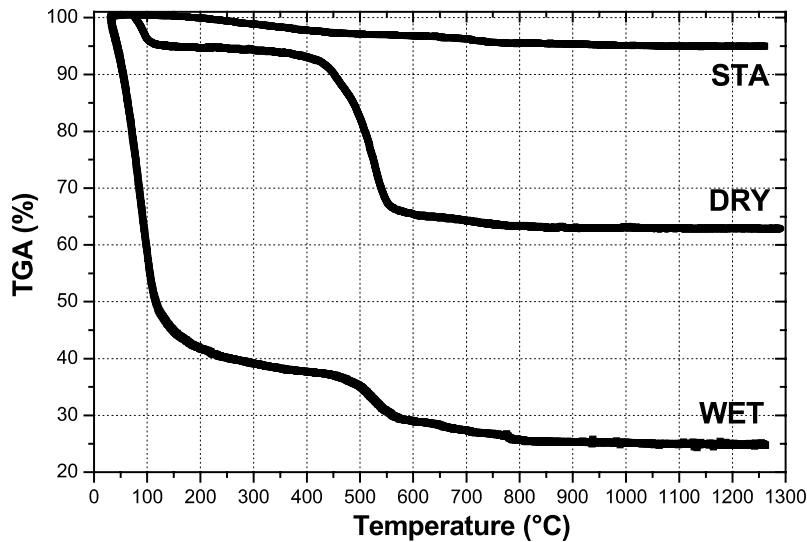


Fig. 5. TGA curves for wet (WET), dry (DRY) and stabilised (STA) S70C30 gel composition.

3.6. XRD traces of stabilised gel-glasses

The XRD study of the stabilised gel glasses confirmed the absence of crystalline phases in the samples (Fig. 6), since no diffraction maxima were observed and only a broad band between 18° and $40^\circ 2\theta$ was detected for all compositions.

3.7. XRD traces of densified gel-glasses

The stabilised gel-glasses were sintered at 1000°C in air for 1 h in a furnace prior to analysis by XRD technique. This analysis was performed to confirm the nature of the final crystallised phases formed during the DTA. The traces obtained are

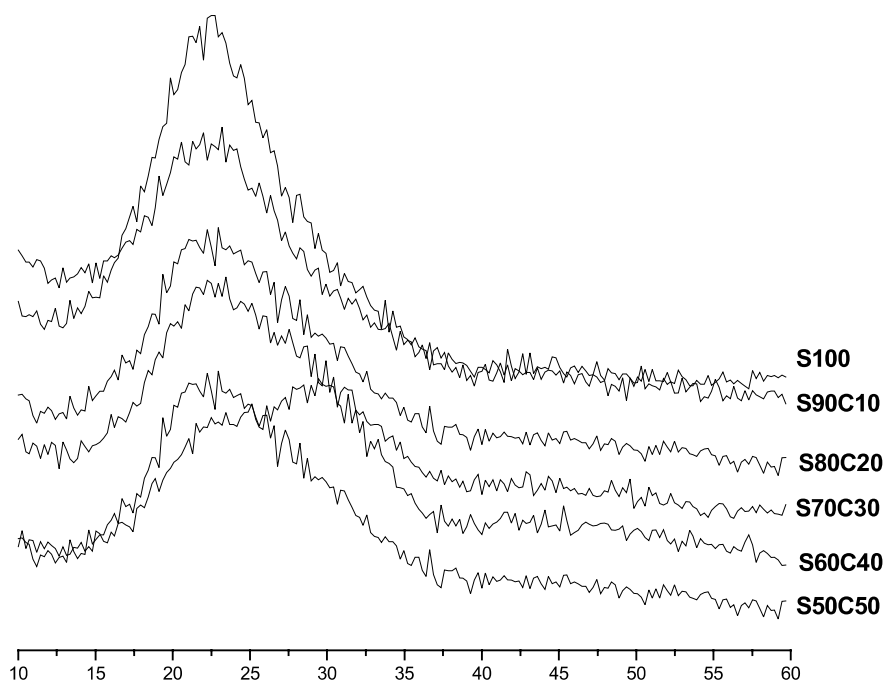


Fig. 6. XRD traces for stabilised gel glasses of different compositions.

illustrated in Fig. 7. Different crystalline phases were identified for glasses with different Ca con-

tent. For glasses with high Ca content (S50C50 and S60C40 glasses), wollastonite was the major

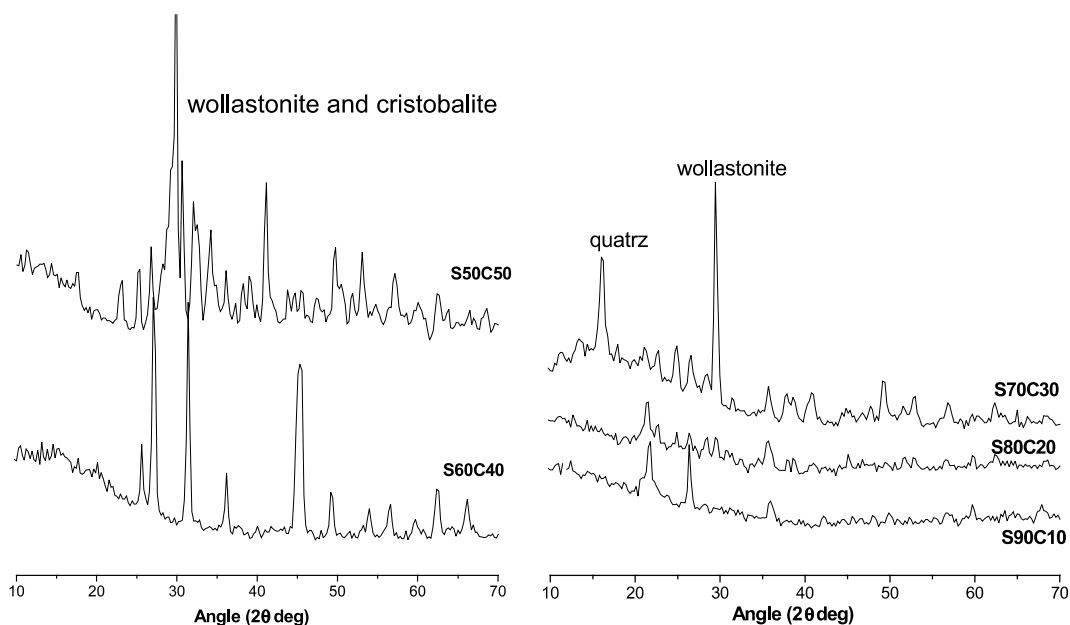


Fig. 7. XRD traces of binary gel glass samples after densification at 1000 °C

phase with traces of cristobalite. For the other glass compositions both α -quartz and wollastonite crystalline phases were present.

3.8. Comparison of FTIR traces at different stages of processing

Fig. 8 compares FTIR traces obtained for a wet, dry and stabilised S70C30 gel-glass sample. The FTIR spectrum for each sample represents an average of 20 scans, which were normalised to the spectrum of a blank KBr pellet. The infrared absorption bands were assigned as follows; bending modes of Si–O–Si and O–Si–O bonds, 460 cm^{-1} ; stretching modes of O–Si–O bonds, 800 cm^{-1} ; Si–O–Ca bonds containing non-bridging oxygen, 950 cm^{-1} ; symmetric stretching vibrations of Si–O–Si bonds, 1080 cm^{-1} ; vibrations of ionic $(\text{NO}_3)^-$, 1380 cm^{-1} ; and vibration of H_2O at 1630 cm^{-1} .

The vibrations due the ionic nitrate are hardly identifiable in the FTIR trace of the stabilised gel-sample and the vibrations due to the silica tetrahedra increase in intensity with progress of the process.

The characteristic band for the O–H stretching vibration was found on all three samples (wet, dry and stabilised). The intensity of the band decreased with the progress of the process. The

presence of traces of water in the stabilised sample along with the absence of any absorption peaks attributable to organic groups may be due to the absorption of atmospheric moisture.

3.9. Comparison of FTIR traces of stabilised gel-glasses of different compositions

Fig. 9 compares the FTIR traces for the stabilised gel-glass samples of all the compositions. As the silica content decreases, the primary peak for the stretching vibrations of the silica tetrahedra (at 1080 cm^{-1}) moved to lower wavenumbers. This band eventually merges with the one for Si–O, terminal non-bridging oxygen (at 950 cm^{-1}) resulting in peak broadening. It was also observed that the intensity of the O–Si–O stretching modes decreased with increasing Ca content.

A possible explanation for this is that in the lower Ca content gels, the number of Q_3 species is high and the Ca cations are held loosely by the nonbridging oxygen atoms. However, when the Ca content is high, the Ca^{2+} ions are incorporated into the silica network as expressed by Eq. (1) [3]:

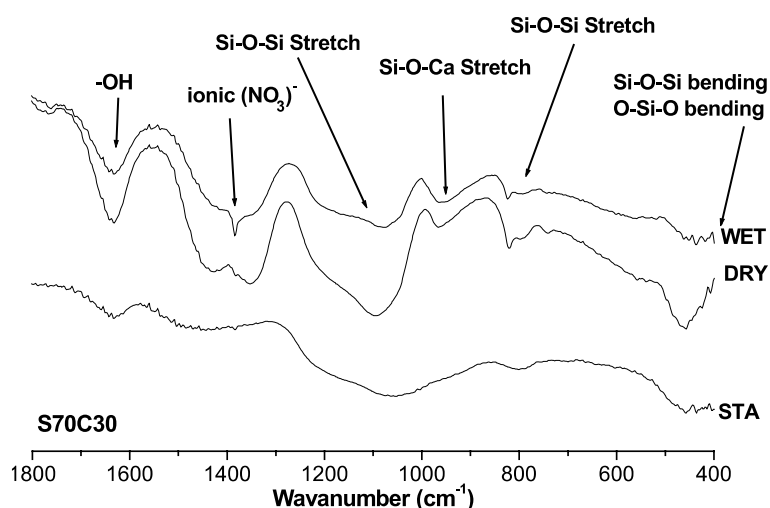
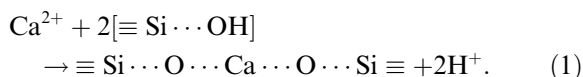


Fig. 8. FTIR spectra of S70C30 gel glass sample after different stages in the process.

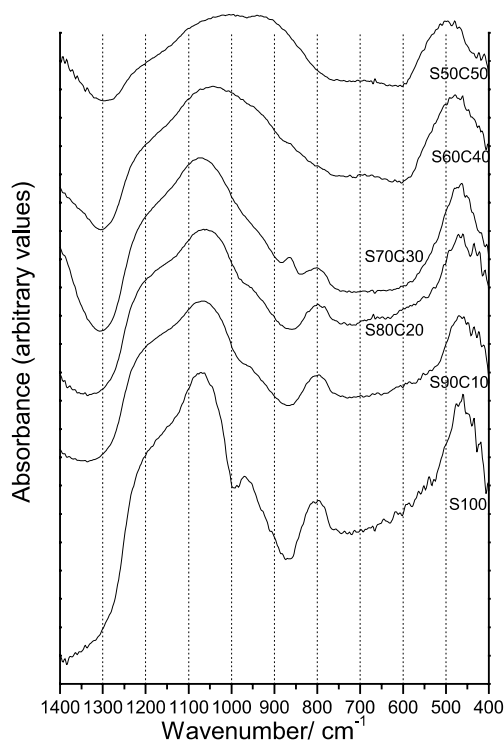


Fig. 9. FTIR spectra of stabilised gel glass samples of different compositions.

3.10. Acid digestion analysis

The results obtained for the six different compositional gel-glasses produced are listed in Table 5 and are graphically illustrated in Fig. 10. The data show that there is negligible difference between the theoretical and experimental values of the SiO_2 content in the stabilised gel-glasses. No other trace compounds (i.e. nitrates) were found indicating that the monoliths were of very high

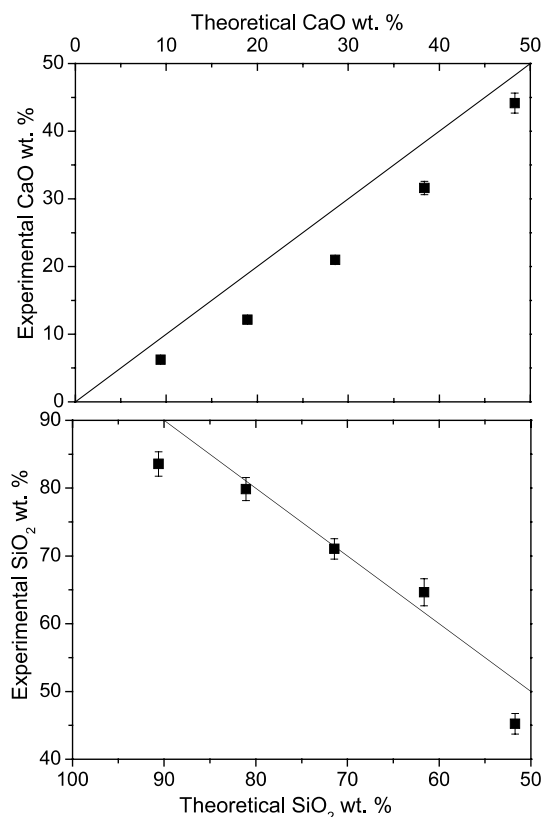


Fig. 10. Graphical representation of Table 5.

purity even when the stabilisation temperature was as low as 600 °C.

3.11. SEM-EDS analysis

Fig. 11(a) illustrates the cross-section of a sample specimen (S80C20 gel glass) mounted between clip holders. EDS analysis was performed

Table 5
Theoretical and experimental oxide content of gel-glasses prepared

Acronym	Experimental (wt%)		Theoretical (wt%)	
	SiO_2	CaO	SiO_2	CaO
S90C10	93.0 ± 1.8	7.0 ± 0.2	90.62	9.38
S80C20	86.8 ± 1.7	13.2 ± 0.3	81.08	18.91
S70C30	77.2 ± 1.5	22.8 ± 0.5	71.43	28.57
S60C40	67.2 ± 2.0	32.8 ± 1.0	61.64	38.36
S50C50	50.6 ± 1.5	49.4 ± 1.5	51.72	48.28

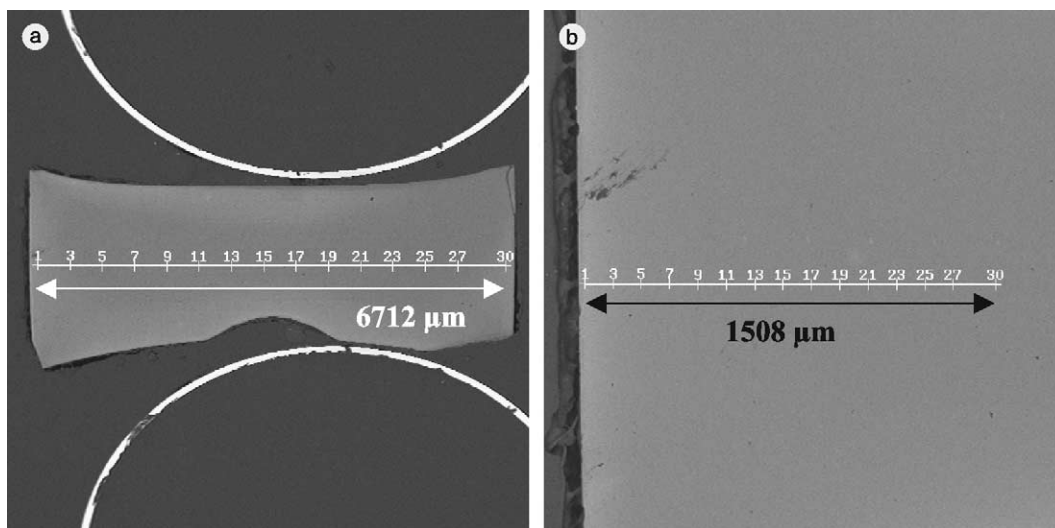


Fig. 11. SEM-EDS image of line scan through a S80C20 specimen disc.

on 30 points along a straight line (i.e. line scan) between the two edges of the specimen as shown in the SEM image (the length of the image is indicated). The results are graphically demonstrated in Fig. 12(a), where the composition (in wt%) is plotted against the distance of the scan point from the edge of the specimen.

The EDS results indicate that the SiO_2 content (in wt%) is higher than both the computed (theoretical) value and the experimental (from acid digestion analysis – see Table 5). The average SiO_2 content in wt% using the EDS analysis is 84.97 ± 1.3 compared to the theoretical value of 81.08 wt% SiO_2 and experimental value of 86.8 wt%. Similarly the average CaO content is 15.03 ± 1.3 wt% compared to that of 18.91 and 13.2 wt% theoretical and experimental values respectively.

The graph indicates that the SiO_2 and CaO content throughout the cross-section of the disc to be fairly constant i.e. it is homogeneous monolith. However, it was observed that the edges of the monolith are slightly rich in CaO. When a similar analysis was undertaken at a higher magnification (Fig. 11(b) – points taken every 50 μm compared to approximately every 230 μm in Fig. 11(a)) it was noted that the slight CaO enrichment was limited

to approximately 200 μm perimeter as seen in Fig. 12(b).

4. Summary and conclusion

Previous literature describes careful preparation of multi-component gels under controlled atmospheres yet with little success. For calcium silicate gel-glasses, a composition limitation of 20-mol% CaO is described for obtaining crack free monoliths. In this paper, the preparation of binary xerogel gel-glasses with up to 50-mol% CaO in its composition has been presented. These gel-glasses are in the form of monoliths of various sizes (discs of 7–25 mm diameter with 2–5 mm thickness; cylinders of 1–3 cm height with 5–10 mm diameter). Crack free monoliths of up to 70–100 pieces have been manufactured during a single batch with yields of up to 90%. This is achieved by controlling the thermal treatment during the aging and drying stages of the sol–gel process. Calcium nitrate tetrahydrate salt is used as the calcium precursor eliminating the need for careful hydrolysis as required with metal alkoxides. The processing technique described above is simple although time consuming.

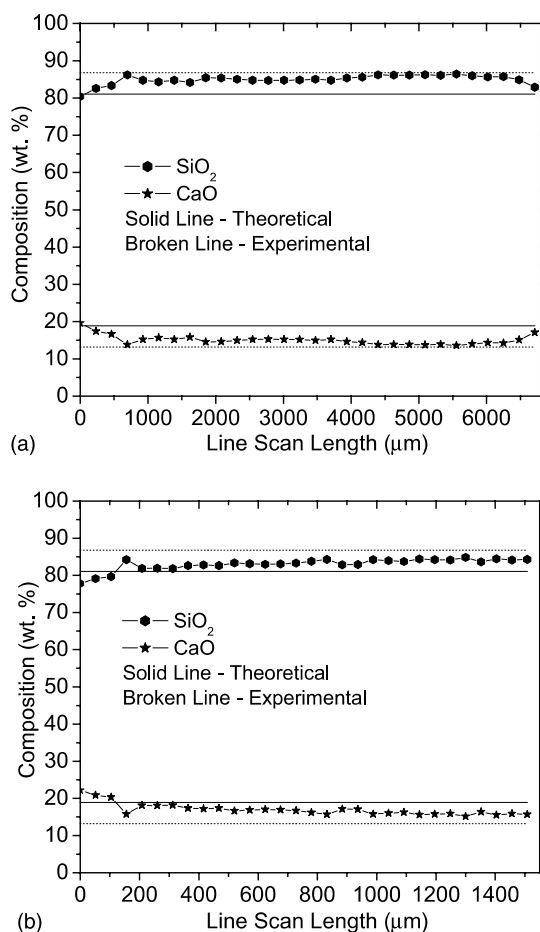


Fig. 12. Compositional analysis using EDS (along the line corresponding to SEM images shown on Fig. 11).

The characterisation results show that the gel-glasses produced consist of only calcium oxide and silica when they are stabilised with no traces of organic matter or nitrates. This is confirmed by the thermal analysis and the FTIR traces at various stages of the process. The gels produced are amorphous glasses (confirmed by XRD) even after stabilisation at 600 °C, with crystallisation temperatures well above 850 °C, estimated using DTA. When sintered at 1000 °C the gel-glasses form crystalline phases in accordance to the CaO–SiO₂ phase diagram.

The acid digestion analysis confirms that the compositions of these gels are similar to that of the

theoretical/computed values. Although the CaO content is always lower than that of the theoretical value, the variation is within 25%. This is attributed to the leaching of cations during the aging and drying stages. The migration of cations is also observed in SEM-EDS analysis where the edges of the monoliths are slightly Ca enriched. However, these gels are fairly homogeneous throughout the monolith.

Thus, it can be concluded that the crack-free CaO–SiO₂ monoliths produced by the sol–gel process described in this paper are homogeneous and of high purity.

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